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Process Optimization for the Extraction of Polyphenol from Malaysian Brown Seaweed, *Sargassum* sp.

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Abstract. The optimum extraction parameters of Malaysian brown seaweed, *Sargassum* sp. were determined by using Response Surface Methodology (RSM). Central Composite Face-centered Design (CCFD) was used to study the influence of extraction temperature (40-60°C), extraction time (60-180 minute) and liquid to solid ratio (20-40 mL/g) on total phenolic content (TPC) of *Sargassum* sp. dried extracts. The statistical analysis of the experiment indicated that extraction temperatue and time had significant effect on *Sargassum* sp. total phenolic contents (TPC). The CCFD showed that quadratic regression models were in good agreement with the experimental results with the coefficients of determination (R^2) of 0.8066 for *Sargassum* sp. TPC. The extraction temperature of 40°C, extraction time of 180 minute and liquid-solid ratio of 40 mL/g were the most efficient for the extraction of polyphenols from *Sargassum* sp. dried powder.

Keywords: extraction condition, total phenolic content, *Sargassum* sp., response surface methodology

INTRODUCTION

Brown seaweeds are rich in bioactive compounds that could potentially be exploited as functional ingredients for both human and animal health applications. The genus *Sargassum*, a kind of brown algae which belonging to the class *Phaeophyceae* have been reported to have a wide range of bioactive activities [7] such as antioxidant, anti-inflammatory, anti-cancer and anti-diabetic [12]. *Sargasum sp*. can be found growing naturally in Pulau Bum Bum, Semporna, Sabah. Three edible brown seaweeds from North Borneo, Malaysia (*Dictyota dichotoma, Sargassum polycyst*um & *Padina* sp.) had been studied by [8] had showing good antioxidant capacity. This indicates that brown seaweed can be a good source of natural antioxidant that can be used as an ingredient in food and beverages product. As such, it is necessary to prepare *Sargassum* sp. dried powder using optimal extraction conditions and further characterize its bioactivity for potential utilization in the functional beverages application in the future.

Extraction is the first important step in isolation of phenolic compounds from plant materials. The main purpose in extraction process is, to provide for the maximum yield of substances and of the highest quality which consist of high concentration of target compounds and antioxidant power of the extracts [11]. Solid-liquid and liquid-liquid extraction techniques is the most commonly used, and has proven to be a reliable and efficient method [3]; [2]. The extraction of polyphenols is demanding due to their chemical structure and their interaction with other food components. Many factors, such as solvent composition, time of extraction, temperature, pH, solid-to-liquid ratio and particle size, may significantly influence the solid–liquid extraction [5].

Optimal extraction conditions which consist of several factors and levels need to be determined. There are two most commonly used for optimization studies, the classical single factor experiments and the response-surface methodology (RSM). Response Surface methodology (RSM) is a collection of statistical and mathematical techniques useful for developing, improving and optimizing processes, in which a response of interest is influenced by several factors (independent variables). A mathematical model, which describes the chemical or biochemical processes can be generate through RSM. Up to now, extraction conditions of total phenolic content from Malaysian brown seaweed (*Sargassum* sp) using RSM has not been reported. In order to determine optimal extraction conditions, this study investigated the effects of a range of experimental parameters including temperature, time and ratio of liquid to solid on the yield of phenolic content.

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MATERIALS AND METHODS

Chemical and Reagents

Gallic acid was purchased from Sigma Sigma-Aldrich (M) Sdn Bhd Chemicals. Sodium carbonate and Folin-Ciocalteau reagent was purchased from Merck (Germany).

Materials and Sample Preparation

Fresh *Sargassum sp*. was purchased from Semporna, Sabah. Seaweed dried powder (SDP) which obtained after drying with an oven at 50°C and grounded in an Ultra Centrifugal Mill (Retsch ZM200) at 0.25 mm diameter sieve to obtain fine powder were used to study the optimal extraction parameters prior to extraction using water. The SDP were refrigerated at -20°C until extraction.

Extraction of Seaweed Dried Powder (SDP)

Approximately 1g of SDP was extracted with different amount of extractant (water) using a temperature controlled shaking water bath at different temperatures for different times. All the extracts were filtered and the filtrates were stored at -20°C until TPC analysis.

Determination of Total Phenolic Content

The TPC in the extract was determined using Folin-Ciocalteau reagent [10] with some modifications. Fresh weight of sample were converted into dry weights on the basis of the moisture content. Each of the aqueous extract (100 µl) was transferred into a test tube and then mixed thoroughly with 0.5 ml Folin- Ciocalteau reagent (prediluted 10-fold with distilled water). After mixing for 5 min, 1.0 ml of 7.5% (w/v) sodium carbonate was added. The mixtures were agitated with a vortex mixer, and then allowed to stand in the dark for a further 120 min at ambient temperature. The absorbance of the non-boiled and boiled water extracts and a prepared blank were measured at 765 nm using a spectrophotometer (UV-vis model 50 Probe). The TPC was expressed as mg gallic acid equivalent (GAE)/g dry weight which was determined from known concentrations of gallic acid standard prepared similarly. Data were reported as a mean ± standard deviation for three replications.

TABLE (1). The uncoded and coded levels of independent variables used for central composite design (CCD).

Independent variables	Variables code	Range and levels (coded)		
		-1	0	+1
extraction temperature (°C)	A	40	50	60
extraction time (minute)	B	60	120	180
liquid-solid ratio (mL/g)	C	20	30	40

Experimental Design for Optimization

An experimental design with 20 experimental runs using three independent variables to optimize the extraction conditions for TPC of *Sargassum sp*. dried powder was carried out. The software Design-Expert (trial version 9, Stat-Ease Inc., Minneapolis, USA) was used for experimental design, regression analysis of the experimental data and quadratic model building. *P*-values of less than 0.05 were considered to be statistically significant. The range and center point values of three independent variables presented in TABLE (1). Response surface methodology (RSM) was used to optimize the extraction parameters with eight factorial point, six axial points and six replicates of the center points according to central composite face-centered design (CCFD) as shown in TABLE (2). Extraction temperature (A), extraction time (B) and liquid to solid ratio (C) were chosen for independent variables. TPC was selected as the response for the combination of the independent variables. Experimental runs were randomized to minimize the effects of unexpected variability in the observed responses.

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TABLE (2). Central composite arrangement for independent variables and their responses to extractions conditions.

Experiment		Independent variables (real values and coded)			TPC (mgGAE/g dw)	
Std. order	Type	Temperature (°C)	Time (minute)	Liquid-solid ratio (mL/g)	observed	predicted
1	Fact	40 (-1)	60 (-1)	20 (-1)	1.07	1.06
2	Fact	60 (+1)	60 (-1)	20 (-1)	0.81	0.79
3	Fact	40 (-1)	180 (+1)	20 (-1)	1.45	1.42
4	Fact	60 (+1)	180 (+1)	20 (-1)	1.03	0.98
5	Fact	40 (-1)	60 (-1)	40 (+1)	0.86	0.88
6	Fact	60 (+1)	60 (-1)	40 (+1)	1.01	1.00
7	Fact	40 (-1)	180 (+1)	40 (+1)	1.27	1.25
8	Fact	60 (+1)	180 (+1)	40 (+1)	1.23	1.21
9	Axial	40 (-1)	120 (0)	30 (0)	1.16	1.21
10	Axial	60 (+1)	120 (0)	30 (0)	0.97	1.05
11	Axial	50 (0)	60 (-1)	30 (0)	0.86	0.89
12	Axial	50 (0)	180 (+1)	30 (0)	1.06	1.18
13	Axial	50 (0)	120 (0)	20 (-1)	0.93	1.03
14	Axial	50 (0)	120 (0)	40 (+1)	1.03	1.06
15	Center	50 (0)	120 (0)	30 (0)	1.13	1.07
16	Center	50 (0)	120 (0)	30 (0)	1.11	1.07
17	Center	50 (0)	120 (0)	30 (0)	1.27	1.07
18	Center	50 (0)	120 (0)	30 (0)	1.02	1.07
19	Center	50 (0)	120 (0)	30 (0)	1.09	1.07
20	Center	50 (0)	120 (0)	30 (0)	1.06	1.07

RESULTS AND DISCUSSION

Analysis of the model

Effect of extraction temperature (40-60°C), time (60-180 minute) and liquid to solid ratio (20-40 mL/g) on TPC were studied by RSM. Experiments were performed according to the central composite face-centered design (TABLE 2). The result of statistical analyses are presented in TABLE (3). The coefficients are related to coded variables. The ANOVA results for modelled responses are reported in TABLE (4).

As summarized in TABLE (3), the Model F-value of 4.64 implies the model is significant. There is only a 1.25% chance that an F-value this large could occur due to noise. Values of "Prob > F" less than 0.05 indicate model terms are significant. In this case, the first-order effects of temperature (*A*) and time (*B*) and the interaction between extraction time and liquid-solid ratio (*AC*) are the significant terms for the responses. However, first-order liquid-solid ratio (*C*) is not the significant model term for the responses but C has to be included to support the hierarchy model. The "Lack of Fit F-value" of 1.51 implies the Lack of Fit is not significant relative to the pure error. There is a 33.08% chance that a "Lack of Fit F-value" this large could occur due to noise. Non-significant lack of fit is good.

Effect of Extraction Conditions on the Content of Total Phenolic in *Sargassum* sp. Dried Extracts

It is obvious from the results in TABLE 3 that there is no main effect from liquid-solid ratio (*C*) within this range. The interaction among independent variables (temperature and time, *AB*; and time and liquid-solid ratio,

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BC) did not have significant effects on TPC (p = 0.2311, p = 0.8919, respectively). A quadratic regression model was made by using coded values from the estimation data:

$$TPC = 1.07 - 0.077A + 0.14B + 0.012C - 0.042AB + 0.099AC + 0.004625BC + 0.063A^2 - 0.035B^2 + 0.022C^2$$

TABLE (3). Analysis of variance (ANOVA) for the TPC from dried *Sargassum* sp. powder.

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	0.37	9	0.041	4.64	0.0125	*significant*
A - temperature	0.060	1	0.060	6.81	0.0261*	
B - time	0.20	1	0.20	22.96	0.0007*	
C- liquid-solid ratio	1.353×10^{-3}	1	1.353×10^{-3}	0.15	0.7033	
AB	0.014	1	0.014	1.63	0.2311	
AC	0.078	1	0.078	8.84	0.0140*	
BC	1.711×10^{-4}	1	1.711×10^{-4}	0.019	0.8919	
A^2	0.011	1	0.011	1.24	0.2914	
B^2	3.385×10^{-3}	1	3.385×10^{-3}	0.38	0.5491	
C^2	1.378×10^{-3}	1	1.378×10^{-3}	0.16	0.7007	
Lack of Fit	0.053	5	0.011	1.51	0.3308	*not significant*
Pure Error	0.035	5	7.011×10^{-3}			

*Significantly difference with $p<0.05$

Statistical parameters obtained from the ANOVA of TPC are given in TABLE (4). The R^2 coefficient gives the proportion of the total variation in the response variable explained or accounted for by the predictors included in the model. In the present study, the predicted R^2 of 0.3407 is in reasonable agreement with the adjusted R^2 of 0.6326 i.e. the difference is less than 0.2. The R^2 coefficient in this study ensured a satisfactory adjustment of the quadratic model to the experimental data. As suggested by Joglekar and May (1987), R^2 should be at least 0.80 for a good fit of a model. In our study, the R^2 values for these response variables are 0.8066, indicating that the model can predict 80.66% of the actual data for TPC and also indicate the adequacy of the applied regression on model. The coefficient of variance (CV) for TPC was found to be (%) 8.76. The CV as the ratio of the standard error of estimate to the mean value of the observed response (as a percentage) is a measure of reproducibility of the model and as a general rule a model can be considered reasonably reproducible if its CV is not greater than 10% [1]. Adequate precision measures the signal to noise ratio. A ratio greater than 4 is desirable. The ratio of 9.454 indicates an adequate signal. Therefore, this model can be used to navigate the design space. FIGURE (1) shows the parity plot of the actual and predicted values for TPC. The plot shows a close fit of the observed values with the predicted ones. The best way of expressing the effect of any extraction parameter on TPC within the investigated experimental range was to generate response surfaces of the model (FIGURE 2).

TABLE (4). Statistical parameters obtained from the ANOVA for the quadratic models.

Variables	TPC (mg GAE/g dw)
R^2	0.8066
R^2 adjusted	0.6326
R^2 predicted	0.3407
S.D.	0.0940
Coefficient of variance (CV) %	8.76
Adequate precision	9.454

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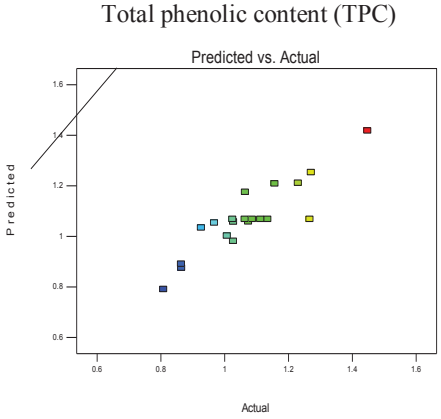


FIGURE 1. Parity plot showing actual versus predicted values for TPC

FIGURE (2) shows the 3D surface plots of the interactive effects of the independent variables corresponding to the response variable. As can be seen from FIGURE 2a, the extraction yield of phenolic content was found high at the temperature of 40°C and liquid-solid ratio was at 20 mL/g. The increasing of extraction temperature and the ratio of liquid to solid will cause the decreasing of TPC [13]. The analysis of the results (TABLE 3) showed that the interaction between the temperature and liquid-solid ratio (*AC*) was the only highly significant factor affecting the phenolic content (p=0.0140). From FIGURE 2b, the TPC was increased when there is an increased in time (up to 180 minute) but decrease slowly when the temperature was increased. The maximum extraction yield of phenolic content was achieved at 180 minute and at the temperature of 40°C. When temperature was constant at 40°C (FIGURE 2C), TPC was also achieve the maximum value at time 180 minute but not significantly influenced by liquid-solid ratio. From the result showed by the 3D surface plots for all the variables, the extraction yield of phenolic content from *Sargassum* sp. dried powder was significantly influenced by the temperature and time (p<0.05) [14]. By decreasing the temperature and increasing time, the total phenolic contents were increased.

Optimization of the Extraction Conditions of *Sargassum* sp. Dried Extracts

Optimization is an essential tool in food engineering for the efficient operation of different processes to yield a highly acceptable product. During optimization of extraction process, several response variables describe the quality characteristics of the obtained extracts. Some of those variables need to be maximized, while others need to be minimized [9]. The optimum conditions for three independent variables, extraction temperature, extraction time and liquid-solid ratio for TPC of *Sargassum* sp. extracts were obtained by using numerical optimization feature of Design Expert 9 (trial version) Software. TABLE (5) shows all the variables and response (TPC) with the respective high-limit and low-limit experimental region that satisfy the criterions defined for the optimum region.

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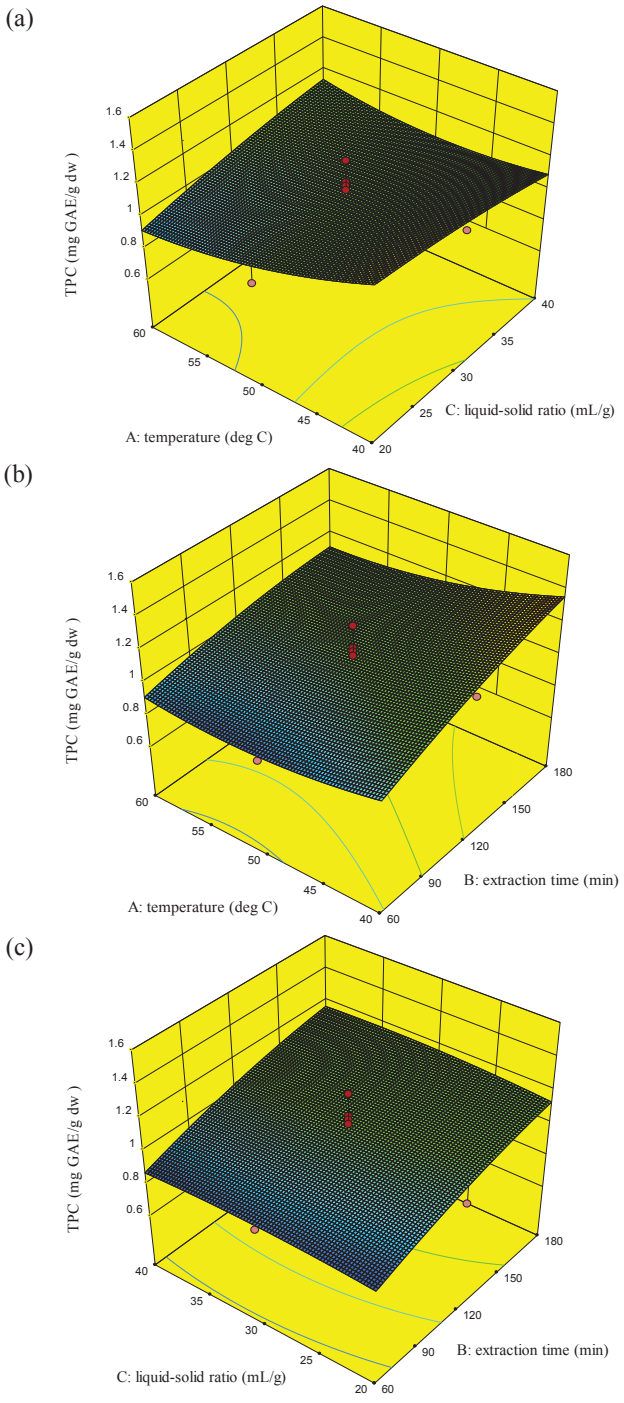


FIGURE 2. Response surface plots showing the effects of investigated parameters on the extraction yield of phenolic and their interactions; (a) time was constant at 120 minute, (b) liquid-solid ratio was constant at 40 mL/g, (c) temperature was constant at 40°C.

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TABLE (5). The preset goal with the constraints for all the independent factors and responses in numerical optimization.

Name	Goal	Experimental design	
		Lower Limit	Upper Limit
Factors			
A: extraction temperature	is in range	40	60
B: extraction time	is in range	60	180
C: ratio of water to sample	maximize	20	40
Response: TPC (mg GAE/g dw)	maximize	0.8081	1.4476

The main goal of this research was to find the best settings for extraction, *i.e.* the most optimal temperature, time and liquid-solid ratio. To optimize the process with two or more output responses, it is helpful to use the concept of desirability function. The desirability function is one of the most widely used methods for optimization of multiple response processes in science and engineering. Desirability ranges from zero to one for any given response. A value of one represent the ideal case, while zero indicates that one or more responses fall outside the desirable limits [4]. By applying desirability function method, the optimum extraction conditions were obtained: temperature of 40°C, time of 180 minute and liquid-solid ratio of 40 mL/g (TABLE 6) with desirability of 0.834

TABLE (6). The optimum conditions and the predicted and experimental value of TPC at the optimum conditions

Factor	Name	Optimum conditions (predicted)	Optimum conditions (experimental)
A	extraction temperature (°C)	40	40
B	extraction time (minute)	180	180
C	ratio of water to sample (mL/g)	40	40
Response	TPC (mg GAE/g dw)	1.2532 ± 0.0938	1.2706 ± 0.1118

Verification of the Model

As shown in TABLE (6), the predicted optimal extraction conditions for TPC of *Sargassum* sp. were found at 40°C extraction temperature, 180 min extraction time and 40 mL/g liquid-solid ratio. In the predicted optimal conditions, the experimental yield of TPC was 1.2532 mg GAE/ g dw. In order to verify the accuracy of the model for predicting maximal yield, we performed actual experiments using the optimized extraction conditions. The experimental values for TPC (1.2706 mg GAE/ g dw) were close to the predicted ones. This indicated that the optimization achieved in the present study was reliable.

CONCLUSIONS

RSM was successfully applied for extraction on TPC of *Sargassum* sp dried powder. During extraction procedures, several factors have effects on the extraction efficiency. RSM is a powerful tool to determine these variables effect that involved in interactions during extraction processes. In this study effects of extraction temperature, time and liquid-solid ratio effects with their interactions was determined. The main significant factors on extraction of phenolic content from *Sargassum* sp. were found as extraction temperature and time. The optimum extraction parameters for polyphenol of *Sargassum* sp. were found at 40°C for 180 min with the ratio water to sample at 40mL/g.

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